

AKHILA ANISH DAS
ROLL NO.150096725016

DBMS — SQL & NOSQL

B.TECH CSE & AI | SEMESTER III
SCHOOL OF FUTURETECH

HIREDesk

JOB PORTAL & RECRUITMENT MANAGEMENT SYSTEM



PROBLEM STATEMENT & OBJECTIVES

RECRUITMENT DATA COMPLEX & SCATTERED

DUPLICATION MAKES SEARCHING DIFFICULT

SKILL MATCHING IS INEFFICIENT

MANAGE RECRUITMENT DATA

REDUCE REDUNDANCY.

MATCH CANDIDATES TO REQUIRED SKILLS.



DATABASE DESIGN

Relational Tables>>

1. Candidate
2. Skill
3. Company
4. JobPosting
5. CandidateSkill
6. JobRequiredSkill
7. Application

Key Design>>

- Primary and Foreign Keys
- Composite keys in junction tables
- Unique constraints
- Normalized relational structure



ER DIAGRAM & RELATIONSHIPS

Main Relationships

- Company → JobPosting: 1 : M
- Candidate → CandidateSkill: 1 : M
- Skill → CandidateSkill: 1 : M
- JobPosting → JobRequiredSkill: 1 : M
- Skill → JobRequiredSkill: 1 : M
- Candidate → Application: 1 : M
- JobPosting → Application: 1 : M



NORMALIZATION

1NF

- Removes multi-valued skill data.
- Skills are stored individually.

2NF

- Composite junction-table keys prevent partial dependency.
- CandidateSkill and JobRequiredSkill use composite primary keys.

3NF

- Separate tables store candidate, skill, company, job, and application information.
- Reduces redundancy and update anomalies.

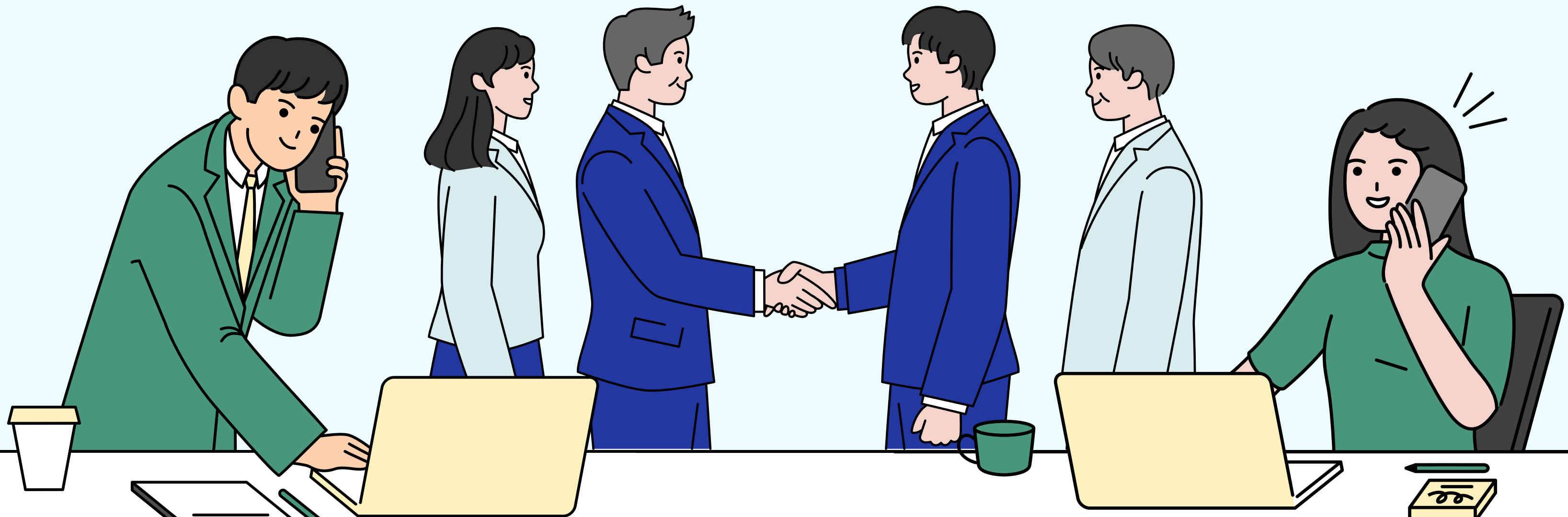
Result

Structured, consistent, and normalized database.



SQL CONCEPTS USED

- JOIN
- Subquery
- GROUP BY
- HAVING
- COUNT()
- DISTINCT
- ORDER BY



INDEXING & APPLICATION TRACKING



Indexes

- JobPosting(Title)
- CandidateSkill(SkillID)
- Faster searching

Applications

- Candidate
- Job
- Applied Date
- Status

Statuses:

Applied • Shortlisted • Interview

RESULTS & CONCLUSION

- Normalized 7-table recruitment database
- Efficient candidate-skill matching
- Application tracking through JOINS
- Faster searches using indexes
- Strong referential integrity

Conclusion>>

HireDesk provides a structured, efficient, and scalable database solution for recruitment management.



THANK YOU

